INTERNSHIP PROJECT REPORT

Game Analytics: Unlocking Tennis Data with SportRadar API

Submitted To

LABMENTIX

Domain

Data Analytics/Business Analytics

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Technologies Used

- Python
- SQLite
- SQL
- Streamlit
- JSON

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to Labmentix for providing us with the opportunity to work on this internship project titled "Game Analytics: Unlocking Tennis Data with SportRadar API." This project has been an excellent learning experience that enabled us to apply theoretical knowledge to a practical data analytics solution.

We are thankful to our trainers and mentors at Labmentix for their constant support, technical guidance, and valuable suggestions throughout the project. Their encouragement helped us understand real-world software development practices, database management, and data visualization techniques.

Working on this project improved our understanding of Python programming, SQL queries, relational databases, JSON data handling, and dashboard development using Streamlit. It also enhanced our teamwork, problem-solving, and analytical thinking skills.

Finally, we would like to thank our family members and friends for their continuous encouragement and motivation throughout the internship.

ABSTRACT

Sports analytics has become an essential component in modern sports management and decision-making. Tennis tournaments generate a large amount of structured data related to competitions, venues, players, rankings, and match statistics. Managing and analyzing this data efficiently requires a reliable database system and an interactive visualization platform.

The objective of this internship project is to design and develop a complete Tennis Game Analytics application capable of storing, organizing, and analyzing tennis-related information. The project demonstrates the complete workflow of data management beginning with structured data generation, database creation, data loading, SQL-based analysis, and dashboard visualization.

The project has been developed using Python, SQLite, SQL, JSON, and Streamlit. Python is used for data generation and processing, SQLite is used as the database management system, SQL is used for data retrieval and analysis, and Streamlit is used to develop an interactive web dashboard.

The application enables users to explore tennis competitions, competitor rankings, venue information, and country-wise player data through an intuitive interface. Users can search competitors, analyze rankings, and explore tournament details using interactive tables and visualizations.

This internship project demonstrates practical implementation of data analytics concepts and highlights how structured sports data can be transformed into meaningful insights using modern analytics tools.

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1. INTRODUCTION

Sports organizations generate large volumes of structured data every day. Tennis tournaments produce information about competitions, players, rankings, venues, countries, and tournament categories. Efficient management of this information is important for organizers, analysts, coaches, and sports enthusiasts.

Traditional methods of storing sports information in spreadsheets make data analysis difficult as the volume of information increases. Database systems provide a structured approach to storing, retrieving, and managing large datasets efficiently.

The objective of this project is to build a Tennis Game Analytics application capable of storing and analyzing tennis competition data through a relational database. The project demonstrates the practical implementation of database design, SQL querying, and interactive dashboard development.

The application stores information related to competition categories, tournaments, venues, player details, and competitor rankings. SQL queries are used to generate meaningful insights from the stored data, while Streamlit provides an interactive interface for users to explore the information.

This project represents a complete data analytics workflow, beginning with structured data generation and ending with visualization through an interactive dashboard.

2.PROBLEM STATEMENT

Sports organizations require an efficient system to manage large volumes of tennis competition data. Managing player information, tournament details, rankings, venues, and competition categories manually is time-consuming and prone to errors.

The objective of this project is to develop a centralized analytics system capable of storing structured tennis data in a relational database and providing analytical insights using SQL queries and interactive dashboards.

The proposed solution enables users to search competitors, analyze rankings, retrieve venue information, and generate useful reports from the stored data.

3.OBJECTIVES

The primary objectives of this project are:

- To design a relational database for storing tennis competition data.
- To generate and organize structured JSON datasets.
- To load structured data into SQLite using Python.
- To perform analytical operations using SQL queries.
- To develop an interactive Streamlit dashboard.

- To provide search functionality for competitors.
- To analyze player rankings and competition information.
- To visualize important insights using charts and tables.
- To demonstrate an end-to-end data analytics workflow.

4.TECHNOLOGY STACK

The Tennis Game Analytics project was developed using modern technologies that support data processing, database management, and interactive dashboard creation. Each technology was selected based on its suitability for handling structured sports data and providing efficient analytical capabilities.

4.1 Python

Python is the primary programming language used for developing this project. It provides a simple syntax, extensive libraries, and strong support for data processing.

Role of Python in the Project

- Generation of structured tennis datasets
- Reading and writing JSON files
- Loading data into the SQLite database
- Executing SQL queries
- Connecting the database with the Streamlit application
- Handling user interactions

Advantages

- Easy to learn and implement
- Large collection of libraries
- Excellent database connectivity
- Fast development process
- Cross-platform compatibility

4.2 SQLite

SQLite is a lightweight relational database management system used to store tennis-related information.

Purpose

The database stores information related to:

- Categories
- Competitions
- Complexes
- Venues
- Competitors
- Competitor Rankings

Advantages

- No installation of database server required
- Portable database file
- Fast query execution
- Supports SQL operations
- Ideal for internship and academic projects

4.3 SQL

Structured Query Language (SQL) is used to retrieve, analyze, and manipulate stored data.

The project uses SQL to:

- Retrieve competitor information
- Display rankings
- Perform joins between tables
- Generate reports
- Filter data
- Sort records
- Aggregate information

4.4 Streamlit

Streamlit is an open-source Python framework used to build the interactive dashboard.

Dashboard Features

- Homepage Dashboard
- Competitor Search
- Rankings Table
- Competition Details
- Venue Information
- Interactive Navigation

Benefits

- Easy web application development
- Interactive UI
- Fast execution
- Python-based development
- Simple deployment

4.5 JSON

JSON (JavaScript Object Notation) is used as the intermediate data format.

The project contains separate JSON files for:

- Categories
- Competitions
- Complexes
- Venues
- Competitors
- Rankings

Python reads these files and imports the data into SQLite.

5.1 SOFTWARE REQUIREMENTS

The project was developed using the following software tools.

Software	Purpose
Python 3.x	Programming Language
SQLite	Database
DB Browser for SQLite	Database Management
Visual Studio Code	Code Editor
Streamlit	Dashboard Development
JSON	Data Storage Format

5.2 HARDWARE REQUIREMENTS

Component	Requirement
Processor	Intel i3 or above
RAM	Minimum 4 GB
Storage	500 MB Free Space
Operating System	Windows 10/11
Internet	Required for package installation

6. PROJECT ARCHITECTURE

The Tennis Game Analytics project follows a structured workflow that transforms raw data into meaningful analytical insights.

Workflow

Generated Tennis Data

▼

JSON Files

▼

Python Data Loader

▼

SQLite Database

▼

SQL Queries

▼

Streamlit Dashboard

▼

Interactive Analytics

Figure 1: Overall Project Architecture

Architecture Explanation

The project begins by generating structured tennis data using Python. The generated data is stored in JSON files corresponding to different entities such as categories, competitions, venues, competitors, and rankings.

The data_loader.py script reads the JSON files and imports the records into the SQLite database while maintaining relationships between the tables.

SQL queries retrieve relevant information from the database. Finally, Streamlit presents this information in the form of an interactive dashboard where users can search, view, and analyze tennis data.

7. DATABASE DESIGN

A relational database was designed to efficiently organize tennis data. The database consists of six interconnected tables.

Database Tables

- Categories
- Competitions
- Complexes
- Venues
- Competitors
- Competitor_Rankings

The design minimizes data redundancy and maintains relationships using primary and foreign keys.

ENTITY RELATIONSHIP OVERVIEW

Categories

|

|

Competitions

Complexes

|

|

Venues

Competitors

|

|

Competitor_Rankings

Figure 2: Simplified Entity Relationship

8. DATABASE TABLES

8.1 Categories Table

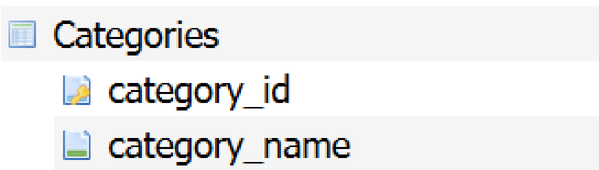
This table stores different categories of tennis competitions.

Columns

- category_id (Primary Key)
- category_name

Purpose

Used to classify competitions into different categories.



8.2 Competitions Table

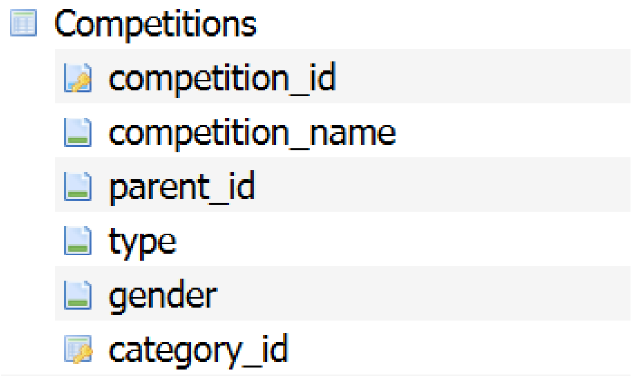
Stores information about tennis competitions.

Columns

- competition_id
- competition_name
- parent_id
- type
- gender
- category_id

Purpose

Maintains details of tournaments and links them to their respective categories.



8.3 Complexes Table

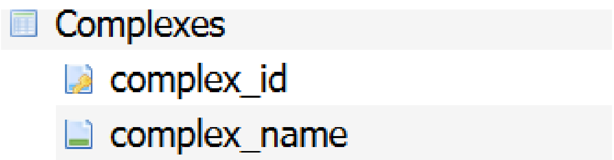
Stores sports complex information.

Columns

- complex_id
- complex_name

Purpose

Represents the sports complexes where tennis venues are located.



8.4 Venues Table

Stores venue information.

Columns

- venue_id
- venue_name
- city_name
- country_name
- country_code
- timezone
- complex_id

Purpose

Stores venue details along with the associated sports complex.

8.5 Competitors Table

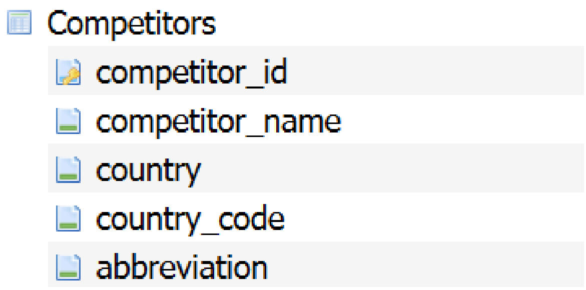
Stores player information.

Columns

- competitor_id
- competitor_name
- country
- country_code
- abbreviation

Purpose

Contains details of tennis players participating in competitions.



8.6 Competitor_Rankings Table

Stores ranking information for competitors.

Columns

- competitor_id
- rank
- movement
- points
- competitions_played

Purpose

Maintains player rankings and performance statistics.

9. IMPLEMENTATION OF THE PROJECT

The Tennis Game Analytics application was developed in a systematic manner. The implementation involved designing the database, generating structured data, loading the data into SQLite, executing SQL queries, and finally creating an interactive dashboard using Streamlit.

The implementation process was divided into the following stages:

1. Database Creation
2. Data Generation
3. Data Loading
4. SQL Query Development
5. Dashboard Development
6. Testing and Validation

This modular approach made the application easy to develop, test, and maintain.

9.1 DATABASE CREATION

The first step of the project was to create the relational database using SQLite. Six tables were created based on the project requirements.

The tables are:

- Categories
- Competitions
- Complexes
- Venues
- Competitors
- Competitor_Rankings

Primary keys and foreign keys were used to maintain relationships between the tables and ensure data integrity.

9.2 DATA GENERATION

Initially, the project was intended to use the SportRadar API. During implementation, API access was unavailable. Therefore, structured datasets were generated using Python while preserving the schema and workflow required by the project.

The generated data included:

- Competition Categories
- Competitions
- Sports Complexes

- Venues
- Competitors
- Competitor Rankings

The generated JSON files followed a structured format that could be directly imported into the SQLite database.

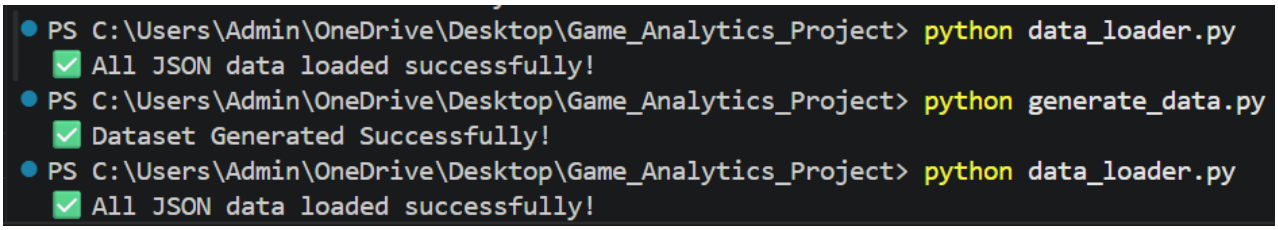
10. DATA LOADING

The data_loader.py program was developed to import all generated JSON files into the SQLite database automatically.

The loader performs the following operations:

- Connects to SQLite
- Clears existing records to avoid duplicates
- Reads JSON files
- Inserts records into the corresponding tables
- Commits the transaction
- Closes the database connection

This automation ensured that the database could be recreated whenever the data changed.



11. PYTHON IMPLEMENTATION

PYTHON MODULES

The project consists of four major Python files.

11.1 database.py

Purpose

This module is responsible for creating the SQLite database and all required tables.

Functions Performed

- Connect to SQLite
- Create database
- Create six tables
- Define primary keys
- Define foreign keys

Importance

It establishes the foundation of the project by preparing the database structure.

11.2 generate_data.py

Purpose

Generates structured tennis datasets.

Functions Performed

- Create realistic sample data
- Generate categories
- Generate competitions
- Generate complexes
- Generate venues
- Generate competitors
- Generate rankings
- Save data as JSON files

Importance

Provides consistent data for analysis and dashboard visualization.

11.3 data_loader.py

Purpose

Loads JSON data into the SQLite database.

Functions Performed

- Read JSON files
- Insert records into tables
- Maintain relationships
- Commit transactions
- Handle duplicate prevention

Importance

Automates the process of populating the database.

11.4 app.py

Purpose

Creates the Streamlit dashboard.

Functions Performed

- Connect to SQLite
- Retrieve data using SQL queries
- Display dashboard metrics
- Display ranking tables
- Search competitors
- Show competitions
- Show venues
- Display charts

Importance

Provides a user-friendly interface for exploring the stored data.

12. SQL QUERIES AND ANALYSIS

SQL queries were used to retrieve meaningful insights from the database. The following analytical queries were implemented.

Note: Due to the large number of SQL queries implemented in this project, only selected queries along with their outputs are presented in this report. The complete SQL script containing all implemented queries is included in the project submission as queries.sql.

QUERY 1. List all competitions with category names

Purpose

To retrieve all competitions along with their respective category names.

SQL Query

```sql
-- 1. List all competitions with category names
SELECT
c.competition_name,
cat.category_name
FROM Competitions c
JOIN Categories cat
ON c.category_id = cat.category_id;
```

OUTPUT

	competition_name	category_name
1	Tennis Championship 1	Grand Slam
2	Tennis Championship 2	WTA Women
3	Tennis Championship 3	ITF Men
4	Tennis Championship 4	ATP 250
5	Tennis Championship 5	ATP Men
6	Tennis Championship 6	ATP Men
7	Tennis Championship 7	WTA 1000
8	Tennis Championship 8	Grand Slam
9	Tennis Championship 9	WTA Women
10	Tennis Championship 10	ATP 500

QUERY 2. Count competitions in each category

Purpose

To determine how many competitions belong to each category.

SQL Query

```
-- 2. Count competitions in each category
SELECT
cat.category_name,
COUNT(c.competition_id) AS Total_Competitions
FROM Categories cat
LEFT JOIN Competitions c
ON cat.category_id = c.category_id
GROUP BY cat.category_name;
```

OUTPUT

	category_name	Total_Competitions
1	ATP 1000	1
2	ATP 250	9
3	ATP 500	6
4	ATP Challenger	6
5	ATP Men	7
6	Grand Slam	2
7	ITF Men	5
8	ITF Women	2
9	WTA 1000	5
10	WTA Women	7

QUERY 3. Display Venues with Complex Name

Purpose

To retrieve venue details together with their corresponding sports complex.

SQL Query

```
-- 8. Venues with Complex Names
SELECT
v.venue_name,
c.complex_name
```

```sql
FROM Venues v
JOIN Complexes c
ON v.complex_id=c.complex_id;
```

OUTPUT

	venue_name	complex_name
1	Stadium 1	Tennis Complex 11
2	Stadium 2	Tennis Complex 3
3	Stadium 3	Tennis Complex 3
4	Stadium 4	Tennis Complex 6
5	Stadium 5	Tennis Complex 3
6	Stadium 6	Tennis Complex 8
7	Stadium 7	Tennis Complex 10
8	Stadium 8	Tennis Complex 3
9	Stadium 9	Tennis Complex 1
10	Stadium 10	Tennis Complex 17

QUERY 4. List Competitors with Rankings

Purpose

To display each competitor along with ranking and points.

SQL Query

```sql
SELECT
c.competitor_name,
r.rank,
r.points
FROM Competitors c
JOIN Competitor_Rankings r
ON c.competitor_id=r.competitor_id;
```

OUTPUT

	competitor_name	rank	points
1	Emma Djokovic	1	6577
2	Andy Sabalenka	2	2009
3	Novak Swiatek	3	8548
4	Andy Raducanu	4	9355
5	Carlos Jabeur	5	8621
6	Maria Osaka	6	3737
7	Paula Badosa	7	1770
8	Jannik Sinner	8	1465
9	Casper Zverev	9	9642
10	Novak Djokovic	10	3074

QUERY 5. Top Ranked Players

Purpose

To identify the highest-ranked tennis competitors.

SQL Query

```
SELECT
c.competitor_name,
r.rank
FROM Competitors c
JOIN Competitor_Rankings r
ON c.competitor_id=r.competitor_id
ORDER BY rank
LIMIT 5;
```

OUTPUT

	competitor_name	rank
1	Emma Djokovic	1
2	Andy Sabalenka	2
3	Novak Swiatek	3
4	Andy Raducanu	4
5	Carlos Jabeur	5

QUERY 6. Competitors by Country

Purpose

To group competitors based on their country.

SQL Query

```
SELECT
country,
COUNT(*) AS Total
FROM Competitors
GROUP BY country;
```

OUTPUT

	country	Total
1	Australia	13
2	France	19
3	Italy	14
4	Spain	24
5	USA	16
6	United Kingdom	14

QUERY 7. Top Players by Points

Purpose

To retrieve competitors with the highest points.

SQL Query

```
SELECT
c.competitor_name,
MAX(points)
FROM Competitors c
JOIN Competitor_Rankings r
ON c.competitor_id=r.competitor_id;
```

OUTPUT

	competitor_name	MAX(points)
1	Roger Sabalenka	11999

13. STREAMLIT DASHBOARD

The Streamlit dashboard was developed to provide an interactive interface for exploring tennis data stored in the SQLite database.

The dashboard includes multiple pages that allow users to access different aspects of the dataset.

13.1 Dashboard Home

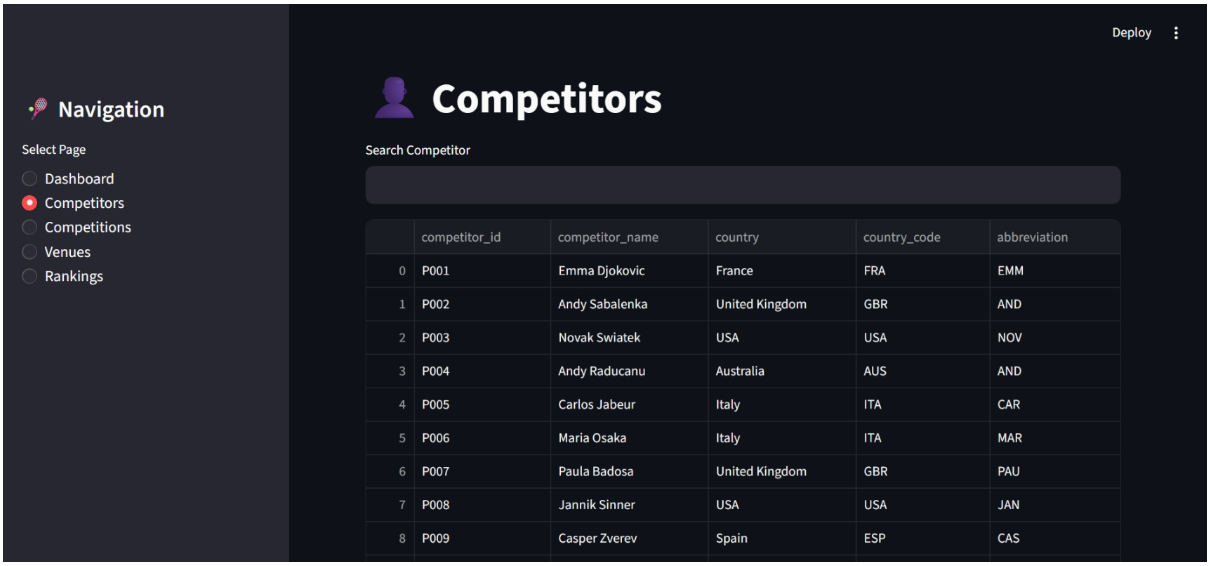
The homepage displays key metrics, including the total number of competitors, competitions, venues, and ranking records. These metrics provide users with an overview of the dataset.

13.2 Competitors Page

The Competitors page displays all competitor information and includes a search feature to quickly find a player by name.

Features

- Competitor list
- Search functionality
- Country information



13.3 Competitions Page

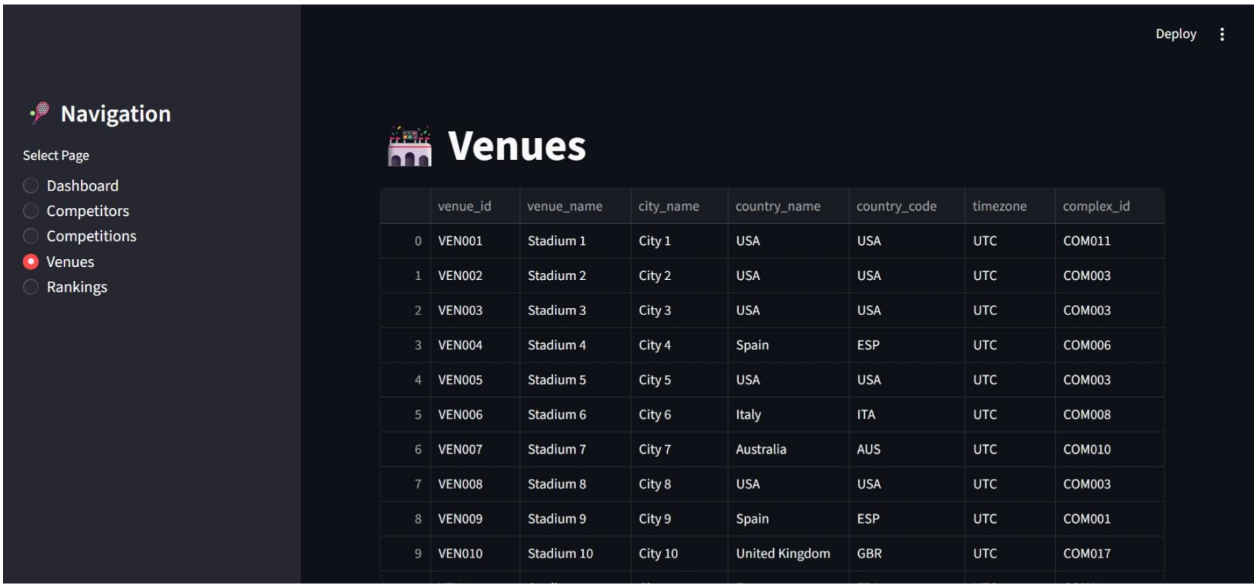
This page displays information about all tennis competitions stored in the database.

Features

- Competition name
- Competition type
- Gender
- Category

13.4 Venues Page

Displays venue information including city, country, timezone, and sports complex.

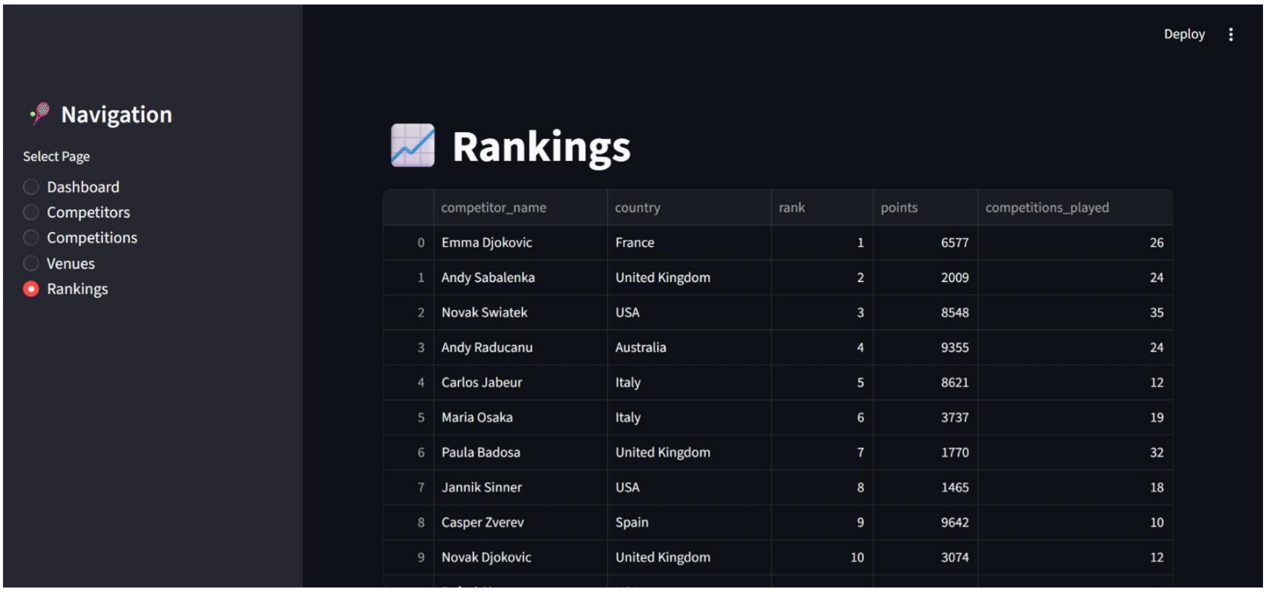


13.5 Rankings Page

The Rankings page displays competitor rankings and a chart showing the top players by points.

Features

- Ranking table
- Points
- Competitions played
- Bar chart visualization



14. RESULTS AND ANALYSIS

The developed application successfully fulfilled the project objectives by integrating database management, SQL analysis, and dashboard visualization into a single system.

Key Results

- Successfully designed and implemented a relational database using SQLite.
- Generated structured tennis datasets following the required schema.
- Loaded all datasets into the database using Python.
- Executed SQL queries to retrieve analytical insights.
- Developed an interactive Streamlit dashboard for data visualization.
- Enabled users to search competitors, view rankings, and explore competition and venue information.

Benefits

- Improved organization of tennis data.
- Faster retrieval of information through SQL queries.
- Interactive visualization for better understanding of the dataset.
- Scalable architecture that can be extended to include live API integration in the future.

15. CHALLENGES FACED

During the development of the Tennis Game Analytics project, our team encountered several technical and practical challenges. These challenges helped us improve our problem-solving abilities and provided valuable practical experience.

15.1 API Accessibility

The original project was designed to retrieve tennis data using the SportRadar API. However, due to API access limitations and the unavailability of a valid API key during the internship, we were unable to fetch live data directly from the API. To overcome this challenge, we developed a Python-based data generation module that produced structured JSON datasets while maintaining the database schema and project workflow.

15.2 Database Design

Designing a relational database with appropriate primary and foreign key relationships required careful planning. Maintaining data consistency across multiple interconnected tables was one of the major challenges during the implementation phase.

15.3 JSON Data Processing

Reading multiple JSON files and importing them into the SQLite database while maintaining referential integrity required careful programming and testing. The team ensured that duplicate records were avoided by clearing existing data before each data loading process.

15.4 Streamlit Dashboard Development

Creating an interactive dashboard that was both user-friendly and informative required multiple iterations. Designing the navigation, displaying analytical data, and presenting charts effectively were important aspects of the dashboard development process.

15.5 Team Coordination

As this was a group internship project, coordinating tasks, managing timelines, and integrating the contributions of all team members required effective communication and teamwork.

16. LEARNING OUTCOMES

The internship project provided us with valuable technical knowledge and practical experience in developing a complete data analytics application. The project significantly enhanced our understanding of database management, Python programming, SQL analysis, and dashboard development.

The key learning outcomes are:

- Developed practical knowledge of Python programming.
- Learned to design and implement relational databases using SQLite.
- Improved SQL querying and database analysis skills.
- Gained hands-on experience in handling structured JSON datasets.
- Learned to automate data loading using Python.
- Developed interactive dashboards using Streamlit.
- Improved analytical thinking and problem-solving abilities.
- Enhanced teamwork and collaboration skills while working on a group project.
- Understood the complete software development lifecycle, from planning to implementation and testing.
- Learned the importance of documentation and project presentation in a professional environment.

This internship also helped bridge the gap between academic concepts and real-world implementation by providing exposure to industry-oriented tools and practices.

17. CONCLUSION

The Game Analytics: Unlocking Tennis Data with SportRadar API project successfully demonstrated the development of a complete sports analytics application using Python, SQLite, SQL, JSON, and Streamlit.

The application effectively stores, manages, and analyzes tennis-related data through a relational database and presents meaningful insights using an interactive dashboard. The project includes database creation, structured data generation, automated data loading, SQL-based analysis, and user-friendly dashboard visualization.

Although live API integration was not possible due to API access limitations, the use of generated structured datasets ensured that all project objectives and functionalities were successfully implemented while following the intended workflow.

The project enhanced our technical knowledge in database design, Python programming, SQL analysis, and dashboard development. It also improved our teamwork, communication, project planning, and documentation skills.

Overall, this internship project provided valuable practical exposure to real-world data analytics concepts and strengthened our confidence in developing end-to-end analytical applications.

18. FUTURE SCOPE

The developed application provides a strong foundation for future enhancements. Several advanced features can be incorporated to increase its functionality and practical usability.

Some possible future improvements include:

- Integration with the live SportRadar API to fetch real-time tennis data.
- Development of player performance prediction models using Machine Learning.
- Addition of tournament statistics and historical match analysis.
- Integration of interactive dashboards using Plotly or Power BI.
- Development of an admin panel for data management.
- Deployment of the application on cloud platforms such as Streamlit Community Cloud, Render, or AWS.
- Implementation of user authentication and role-based access.

- Support for multiple sports such as football, cricket, basketball, and badminton.
- Advanced data visualization using interactive charts and filters.
- Mobile-friendly dashboard interface for better accessibility.

These enhancements would improve the analytical capabilities and make the application suitable for real-world sports analytics environments.

Project Repository

The complete source code, project documentation, SQL scripts, and dashboard files for this internship project are available in the GitHub repository below.

GitHub Repository:

https://github.com/Jiya1001/Game-Analytics-Unlocking-Tennis-Data-with-SportRadar-API-.git

19. REFERENCES

The following resources were referred to during the development of this internship project:

1. SportRadar API Documentation – Project reference for the required data structure and workflow.
2. Python Official Documentation – https://docs.python.org/
3. SQLite Official Documentation – https://www.sqlite.org/docs.html
4. Streamlit Official Documentation – https://docs.streamlit.io/
5. Pandas Documentation – https://pandas.pydata.org/docs/
6. Matplotlib Documentation – https://matplotlib.org/stable/index.html
7. JSON Documentation – https://www.json.org/
8. DB Browser for SQLite Documentation – https://sqlitebrowser.org/